Labs: Fingerprint Assignment
Image Based Biometry

Asst. Prof. Dr. Žiga Emeršič,
Asst. Dr. Matej Vitek,
Prof. Dr. Peter Peer
ziga.emersic@, matej.vitek@ peter.peer@fri.uni-lj.si

Faculty of Computer and Information Science
University of Ljubljana

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NBIS
mindtct
bozorth
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NBIS Tools Summary

`mindtct` Extracts minutiae points (ridge endings and bifurcations) from fingerprint images. It enhances the image, segments the fingerprint area, estimates ridge orientation, and outputs a `.xyt` template containing minutiae coordinates, directions, and types.

`bozorth3` Compares two minutiae templates and computes a similarity score indicating whether they likely originate from the same fingerprint. It builds geometric relationships between minutiae, constructs a compatibility graph, and finds the largest set of consistent pairings.

`nfiq` Assesses fingerprint image quality on a scale from 1 (best) to 5 (worst). It computes features such as ridge clarity, contrast, and minutiae density, and uses a pre-trained neural network to assign a quality class.

`pcasys` Classifies fingerprints into global pattern types (Arch, Left Loop, Right Loop, Scar, Tented Arch, Whorl). It extracts ridge orientation features, reduces dimensionality with PCA, and assigns a class label based on statistical classification.

mindtct – Minutiae Extraction

- Purpose: Extracts minutiae points (ridge endings and bifurcations) from fingerprint images.
- How it works:
  - Enhances and segments the fingerprint area.
  - Estimates local ridge orientation.
  - Detects minutiae and removes false points.
- Output: `.xyt` file with coordinates, orientation, and type of minutiae.

bozorth3 – Minutiae Matching

- Purpose: Compares two minutiae templates and computes a similarity score.
- How it works:
  - Builds geometric relationships between minutiae.
  - Constructs a compatibility graph between two templates.
  - Finds the largest set of consistent pairings to calculate match score.
- Output: Numeric similarity score (higher = more likely same fingerprint).



nfiq – Quality Assessment

- Purpose: Evaluates fingerprint image quality on a 1–5 scale.
- How it works:
  - Computes image features (ridge clarity, contrast, minutiae density).
  - Uses a trained neural network to assign a quality class.
- Output: Quality score (1 = excellent, 5 = poor).

pcasys – Pattern Classification

- Purpose: Classifies fingerprints into global types (A, L, R, S, T, W).
- How it works:
  - Extracts ridge-orientation features and reduces them with PCA.
  - Applies a classifier to assign the most probable pattern class.
- Output: Pattern label and confidence score.